EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

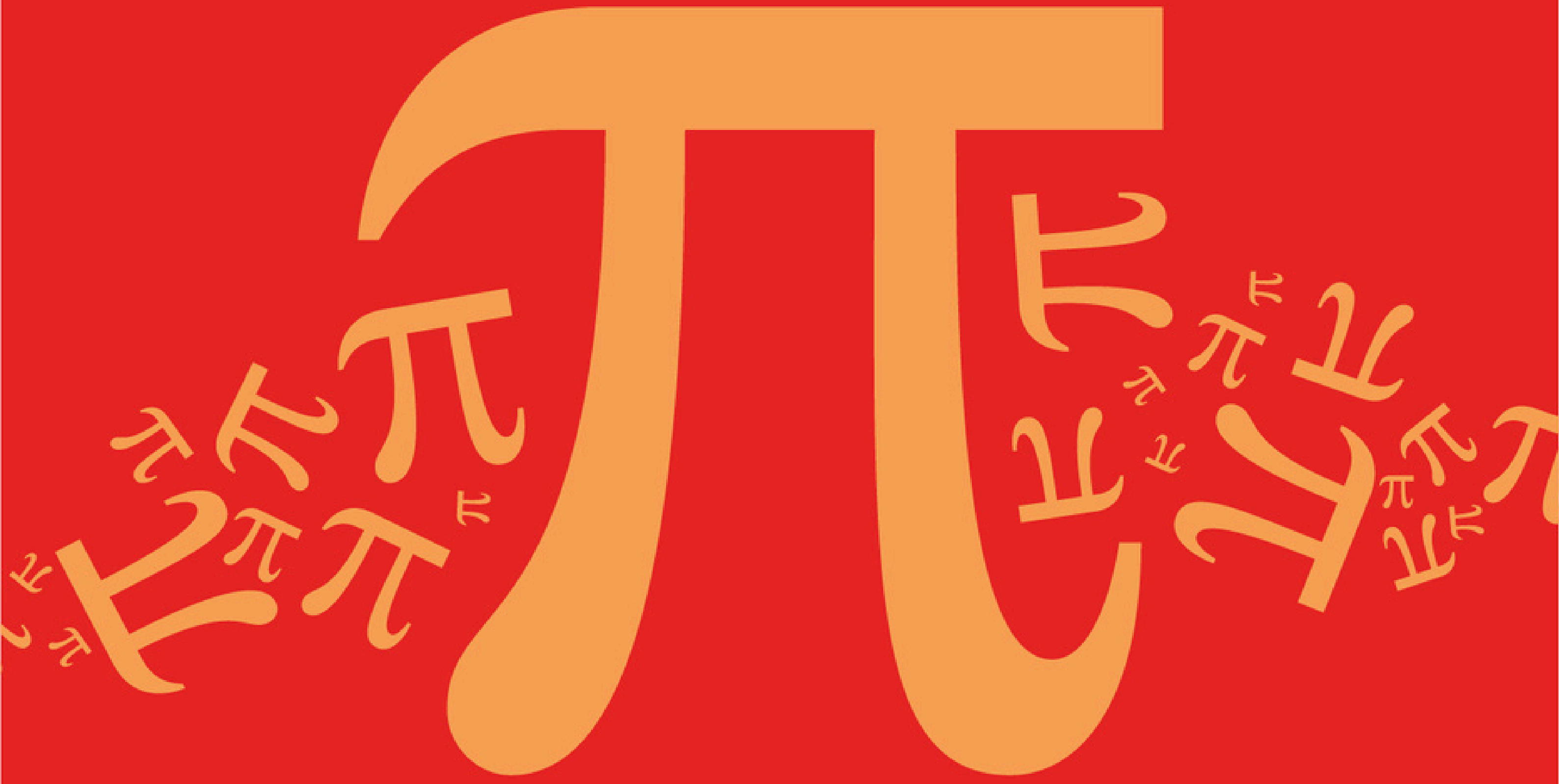
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic **C: Matrices**

Sub topic **C11: Diagonalisation**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**C: Matrices**
Sub topic	**C11: Diagonalisation**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ______________________

Date : ______________________

Time : ______________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

$\mathbf{M} = \begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix}$

(a) Given that 4 is an eigenvalue of **M**, find a corresponding eigenvector.

(3)

(b) Given that **MU** = **UD**, where **D** is a diagonal matrix, find possible matrices for **D** and **U**.

(8)

(Total 11 marks)

Q2.

The matrix **M** is defined by

$$\mathbf{M} = \begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix}$$

where a is a real number. The distinct eigenvalues of **M** are λ_1, λ_2 and λ_3 with corresponding eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.

(a) Given that $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, find λ_1.

(2)

(b) Given that $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, find the value of **a**.

(3)

(c) Given that $\lambda_3 = -6$, find a possible eigenvector $\mathbf{v}_3$.

(3)

(d) The matrix **M** can be expressed as $\mathbf{UDU}^{-1}$, where **D** is a diagonal matrix.

Write down possible matrices **D** and **U**.

(3)

(Total 11 marks)

Q3.

(a) (i) Find the eigenvalues and corresponding eigenvectors of $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix}$.

(6)

(ii) Hence write down each of the matrices **U**, **D** and $\mathbf{U}^{-1}$ such that $\mathbf{A} = \mathbf{UDU}^{-1}$, where **D** is a diagonal matrix.

(4)

(b) A 2 × 2 matrix **M** has distinct real eigenvalues γ and μ, with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

(i) By considering the diagonalised form of **M**, determine the eigenvalues of $\mathbf{M}^3$.

(2)

(ii) Write down the eigenvectors of $\mathbf{M}^3$

(1)

(Total 13 marks)

Q4.

The transformation T is represented by the matrix **M** with diagonalised form

$$\mathbf{M} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$$

where $\mathbf{U} = \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} 27 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) (i) State the eigenvalues, and corresponding eigenvectors, of **M**.

(4)

(ii) Find a cartesian equation for the line of invariant points of T.

(2)

(b) Write down $\mathbf{U}^{-1}$, and hence find the matrix **M** in the form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where a, b, c and d are integers.

(5)

(c) By finding the element in the first row, first column position of $\mathbf{M}^n$, prove that

$$4 - 3^{3n+1} + 1$$

is a multiple of 13 for all positive integers n.

(5)

(Total 16 marks)

Q5.

The 2 × 2 matrix **M** has an eigenvalue 3, with corresponding eigenvector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and a second eigenvalue –3, with corresponding eigenvector $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

The diagonalised form of **M** is $\mathbf{M} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$.

(a) (i) Write down suitable matrices **D** and **U**, and find $\mathbf{U}^{-1}$.

(4)

(ii) Hence determine the matrix **M**.

(3)

(b) Given that n is a positive integer, use the result $\mathbf{M}^n = \mathbf{U}\,\mathbf{D}^n\,\mathbf{U}^{-1}$ to show that:

(i) when n is even, $\mathbf{M}^n = 3^n\,\mathbf{I}$;

(ii) when n is odd, $\mathbf{M}^n = 3^{n-1}\,\mathbf{M}$.

(6)

(Total 13 marks)

Q6.

The matrix **T** has eigenvalues 2 and –2, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ respectively.

(a) Given that $\mathbf{T} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$, where **D** is a diagonal matrix, write down suitable matrices **U**, **D** and $\mathbf{U}^{-1}$.

(3)

(b) Hence prove that, for all **even** positive integers n,

$$\mathbf{T}^n = \mathrm{f}(n)\,\mathbf{I}$$

where $\mathrm{f}(n)$ is a function of n, and **I** is the 2×2 identity matrix.

(5)

(Total 8 marks)

Q7.

A plane transformation is represented by the 2×2 matrix **M**. The eigenvalues of **M** are 1 and 2,

with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ respectively.

(a) State the equations of the invariant lines of the transformation and explain which of these is also a line of invariant points.

(3)

(b) The diagonalised form of **M** is $\mathbf{M} = \mathbf{U}\,\mathbf{D}\,\mathbf{U}^{-1}$, where **D** is a diagonal matrix.

(i) Write down a suitable matrix **D** and the corresponding matrix **U**.

(2)

(ii) Hence determine **M**.

(4)

(iii) Show that $\mathbf{M}^n = \begin{bmatrix} 1 & f(n)-1 \\ 0 & f(n) \end{bmatrix}$ for all positive integers n, where $f(n)$ is a function of n to be determined.

(3)
(Total 12 marks)

Q8.

(a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

(6)

(b) (i) Write down a diagonal matrix **D**, and a suitable matrix **U**, such that

$$\mathbf{X} = \mathbf{UDU}^{-1}$$

(2)

(ii) Write down also the matrix $\mathbf{U}^{-1}$.

(1)

(iii) Use your results from parts (b)(i) and (b)(ii) to determine the matrix $\mathbf{X}^5$ in the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where a, b, c and d are integers.

(3)
(Total 12 marks)

Q9.

For real numbers a and b, with $b \neq 0$ and $b \neq \pm a$, the matrix

$$\mathbf{M} = \begin{bmatrix} a & b+a \\ b-a & -a \end{bmatrix}$$

(a) (i) Show that the eigenvalues of **M** are b and $-b$.

(3)

(ii) Show that $\begin{bmatrix} b+a \\ b-a \end{bmatrix}$ is an eigenvector of **M** with eigenvalue b.

(2)

(iii) Find an eigenvector of **M** corresponding to the eigenvalue $-b$.

(2)

(b) By writing **M** in the form $\mathbf{UDU}^{-1}$, for some suitably chosen diagonal matrix **D** and corresponding matrix **U**, show that

$$\mathbf{M}^{11} = b^{10}\mathbf{M}$$

(7)
(Total 14 marks)

Q10.

The matrix **M** is given by $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix}$, where k is a constant. It is given that the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of **M**.

(a) Show that $k = 5$ and find the eigenvalue corresponding to the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(3)

(b) Determine the second eigenvalue of **M** and find a corresponding eigenvector.

(5)

(c) Write down matrices **U** and **D**, having integer elements, such that **M** can be expressed in the diagonalised form $\mathbf{M} = \mathbf{UDU}^{-1}$.

(2)

(d) Write down the matrix $\mathbf{U}^{-1}$.

(1)

(e) The matrix $\mathbf{M}^{2n}$, for positive integers n, is such that

$$\mathbf{M}^{2n} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

for integers a, b, c and d. Show that

$$a = p \times 4^n + q \times 81^n$$

where p and q are rational numbers to be determined.

(4)
(Total 15 marks)